

Supplementary Information

Table of Contents

Supplementary Text

The CSPdb

NMR ring-current contributions to chemical shift perturbations

NMR electric-field induced chemical shift perturbations

Case Studies Figure Format

Supplementary Tables

SI Table S1. Data for all receptors in this study

SI Table S2. Hydrolase receptors

SI Table S3. Transferase receptors

SI Table S4. All- α receptors

SI Table S5. All- β receptors

SI Table S6. $\alpha + \beta$ ($a+b$) receptors

SI Table S7. BET-ET domain receptors

SI Table S8. Ubiquitin domain receptors

SI Table S9. Dissimilar apo/olo experimental conditions

Supplementary Figures

SI Fig. S1. Confusion histograms — hydrolases

SI Fig. S2. Confusion histograms — transferases

SI Fig. S3. Confusion histograms — all- α proteins

SI Fig. S4. Confusion histograms — all- β proteins

SI Fig. S5. Confusion histograms — $\alpha + \beta$ ($a+b$) proteins

SI Fig. S6. Confusion histograms — BET-ET domains

SI Fig. S7. Confusion histograms — ubiquitin domains

SI Fig. S8. Dissimilar apo/olo conditions ($n = 108$)

SI Fig. S9. Figure 3 for same-author/same-study subset ($n = 32$)

SI Fig. S10. Confusion histograms calculated with Eqn. 2 ($n = 121$)

SI Fig. S11. Confusion histograms calculated with Eqn. 3 ($n = 101$)

SI Fig. S12. F1 for Eqn. 1 CSPs vs Eqn. 2 CSPs ($n = 121$)

SI Fig. S13. Confusion by closest interchain N–N distance ($n = 135$)

SI Fig. S14. Confusion by closest interchain any-atom distance ($n = 135$)

SI Fig. S15. 1D H/N/C α /H α F1 boxplots ($n = 101$)

SI Fig. S16. FP/(TP+FP) distributions under max(cleaned mean, 0.05 ppm) ($n = 83/52/135/243$)

SI Fig. S17. CSP significance and z-score vs interchain distance ($n = 135$)

SI Fig. S18. Case study — 2FIN / apo BMRB 6809

SI Fig. S19. PDB Advanced Search Settings

SI Fig. S20. Buffer pH/temperature threshold sweep

SI Fig. S21. Ideal N/H and HA/CA chemical-shift offset grids ($n = 135/101$)

SI Fig. S22. Terminal-anchor vs global offsets (HSQC overlays and CSP bars)

SI Fig. S23. CSP significance threshold distribution ($n = 135$)

SI Fig. S24. Figure 3 under alternate CSP significance thresholds ($n = 135$)

SI Fig. S25. Apo–apo control studies

S25A. BMRB 52080 / BMRB 52079

S25B. BMRB 28070 / BMRB 28071

S25C. BMRB 34000 / BMRB 34001

S25D. BMRB 34394 / BMRB 6354

S25E. BMRB 17769 / BMRB 51725

SI Fig. S26. F1 vs MCC ($n = 135$ buffer-similar)

Supplementary Equations

SI Equation S1. Combined HN chemical-shift perturbation

SI Equation S2. Combined HN+C α chemical-shift perturbation

SI Equation S3. Combined H α +C α chemical-shift perturbation

References

References cited in this Supplementary Information

All Case Studies

Case studies for all $n = 243$ CSPdb targets (one page per target), appended after the References